

Trigonometry

8

SYLLABUS 3.5–3.11

Your voice is a wave. So is the light from your screen, the signal carrying this page to your device, the current in the wall socket, the tide in Tokyo Bay, and the beating of your own heart.

They do not look alike. A heartbeat is a sharp spike, a spoken vowel is jagged, a tide is a slow rise and fall over twelve hours. In 1822 Joseph Fourier made a startling claim: a repeating signal, however jagged, can be built up from pure sine waves. Choose enough of them, with the right heights and the right timings, and the shape appears. Your phone uses this idea many times a second to compress your voice.

So the same curve describes all of them. Where does it come from?

Watch one seat on a turning Ferris wheel. As the wheel turns steadily, the seat rises, slows near the top, falls, slows near the bottom, and rises again. Plot its height against time and the sine curve appears. The seat is a point moving round a circle, and its height is the vertical coordinate of that point. The circle and the wave are the same motion seen two ways: one going round, one going along.

That is why this chapter starts with a circle. In Ch. 7 sine and cosine were ratios inside a right-angled triangle, so the angle had to be acute. On the circle they become functions of *any* angle, and they repeat forever.

By the end of this chapter you will be able to read exact values of sine, cosine and tangent from the unit circle. You will use identities to rewrite one expression as another, sketch and transform the graphs of these functions, and work with the reciprocal and inverse functions. You will also solve trigonometric equations, where a single value of the sine or cosine is shared by two different angles, so answers usually have to be looked for in pairs.

8.1 The Unit Circle

How do you find $\sin 120^\circ$? Or $\cos 360^\circ$, or $\tan(-30^\circ)$?

You already have a definition, and it is SOHCAHTOA. In a right triangle,

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Now try to draw the triangle. For 120° there is none: the three angles add to 180° , one of them is already the right angle, so the other two must each be under 90° . For 360° there is none either, and -30° is worse, because a triangle has no negative angles. Under SOHCAHTOA these three expressions have no meaning at all.

Yet your calculator answers all three without complaint, and a turn of 120° is an ordinary thing to make. So the definition has to widen. Whatever replaces it must agree with SOHCAHTOA wherever SOHCAHTOA works, and keep working where it does not.

The unit circle does exactly that. Take the circle of radius 1 centred at the origin, and a point P on it at angle θ , measured anticlockwise from the positive x -axis.

For an acute θ , drop a perpendicular from P to the x -axis. This makes a right triangle whose hypotenuse is the radius, so the hypotenuse is 1. The old ratios then read

$$\sin \theta = \frac{\text{opposite}}{1} = \text{opposite}, \quad \cos \theta = \frac{\text{adjacent}}{1} = \text{adjacent},$$

and those two sides are exactly the coordinates of P . So for acute angles the new definition agrees with the old one.

The new form never mentions a triangle. A point can sit anywhere on the circle, at any angle, and it always has coordinates. That is what extends the definition to every real θ .

DEF On the unit circle, for a point P at angle θ :

$$\cos \theta = x\text{-coordinate of } P, \quad \sin \theta = y\text{-coordinate of } P.$$

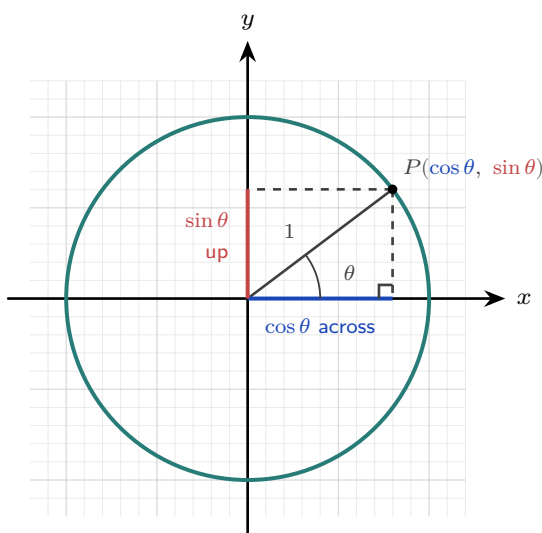


Figure 8.1 On a circle of radius 1 the hypotenuse is 1, so the two shorter sides *are* the coordinates of P .
The cobalt side, measured across, is $\cos \theta$; the scarlet side, measured up, is $\sin \theta$.

The tangent is the gradient of the line OP , so $\tan \theta = \frac{\text{rise}}{\text{run}} = \frac{\sin \theta}{\cos \theta}$. It is undefined whenever $\cos \theta = 0$.

KEY · IB $\tan \theta = \frac{\sin \theta}{\cos \theta}$.

That word *gradient* is not accidental. The line OP passes through the origin, and its gradient is $\tan \theta$, so its equation is $y = x \tan \theta$. This ties the tangent to the straight lines of Ch. 4: a gradient and an angle are two names for the same thing, and the tangent converts between them. A line of gradient 1 makes an angle of $\frac{\pi}{4}$ with the positive x -axis, because $\tan \frac{\pi}{4} = 1$.

KEY A line through the origin making an angle θ with the positive x -axis has equation

$$y = x \tan \theta.$$

Read forwards it gives the gradient from the angle; read backwards, $\theta = \arctan(\text{gradient})$ gives the angle from the gradient.

8.1.1 Exact values

A few angles give exact coordinates, and they come from the 30–60–90 and 45–45–90 triangles.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Only three numbers appear in the shaded row, and they are all that has to be memorised:

$$\frac{1}{2}, \quad \frac{\sqrt{2}}{2}, \quad \frac{\sqrt{3}}{2}.$$

The cosine row is those same three in reverse order. That is not a coincidence. Since $\cos \theta = \sin(\frac{\pi}{2} - \theta)$, reading the sine row backwards reads the cosine row forwards. The tangent row needs nothing new at all: $\tan \theta = \frac{\sin \theta}{\cos \theta}$, so divide one row by the other. At $\theta = \frac{\pi}{2}$ the cosine is zero, which is why the tangent has no value there.

The first-quadrant values repeat into the other three quadrants with signs changed. Collecting them all onto one circle gives the diagram to learn.

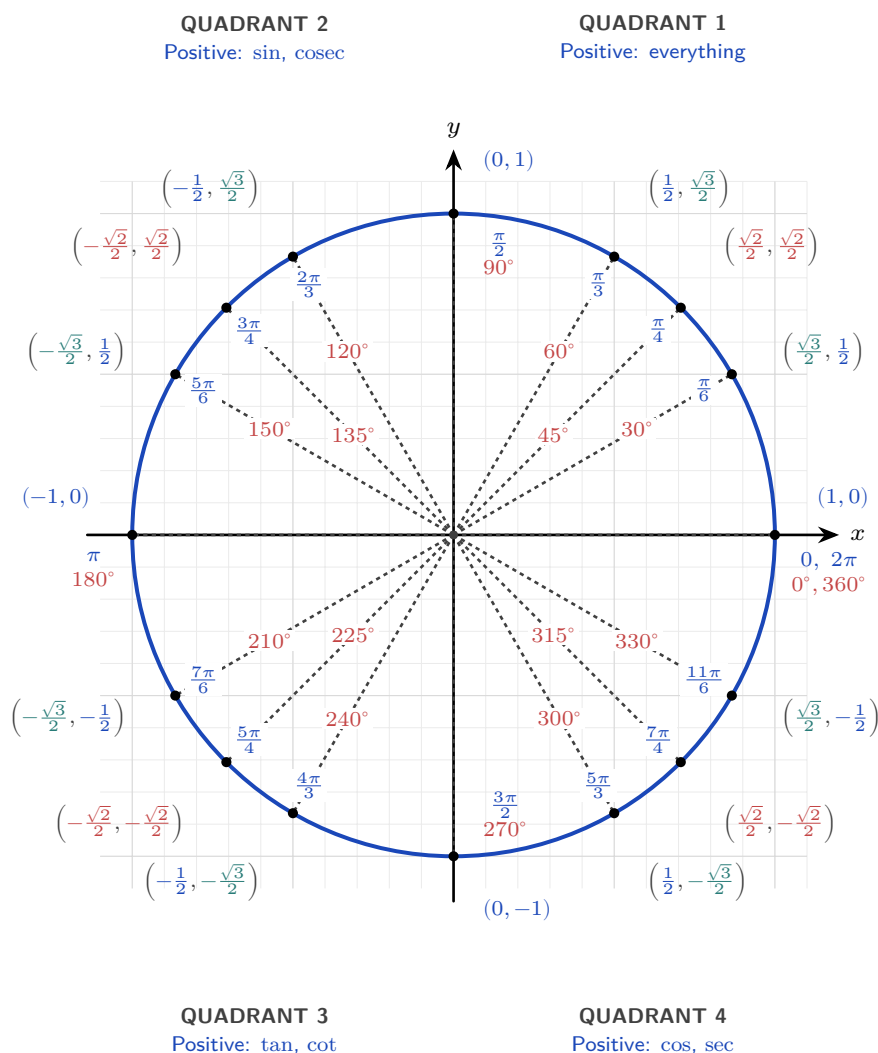


Figure 8.2 The unit circle with every special angle. Each spoke carries the angle twice: in cobalt on the outer ring in radians, and in scarlet on the inner ring in degrees. Outside the circle is the point $(\cos \theta, \sin \theta)$, with each coordinate coloured by its size: cobalt for $\frac{1}{2}$, scarlet for $\frac{\sqrt{2}}{2}$, teal for $\frac{\sqrt{3}}{2}$. To read a value, find the angle and take the matching coordinate. The corner labels say which functions are positive in each quadrant; sec, cosec and cot are the reciprocal functions of §8.4.

The colours make the point. Only three numbers appear anywhere on the circle, and every coordinate on it is one of those three with a sign attached. Add the four points on the axes and the diagram is complete.

8.1.2 Symmetry and the sign of each function

Since $\cos \theta$ and $\sin \theta$ are the coordinates of P , their signs are just the signs of x and y in each quadrant. Nothing needs to be memorised separately: in the second quadrant x is negative and y is positive, so $\cos \theta < 0$ and $\sin \theta > 0$, and $\tan \theta = \frac{\sin}{\cos}$ is negative.

Figure 8.2 records the answer at each corner. Reading the four quadrants anticlockwise from the first, the functions that stay positive are all three, then sine alone, then tangent alone, then cosine alone.

The mnemonic **All Students Take Calculus** is the usual way to remember that order. Use it as a check, not as the reason. The reason is always the sign of the coordinates, and if you can picture the point you never need the letters at all.

Each symmetry of the circle now gives an identity, and each one is a statement about where a reflected point lands.

Reflect in the y -axis. The point at angle $\pi - \theta$ is the mirror image, across the y -axis, of the point at angle θ . Reflecting in the y -axis negates the x -coordinate and leaves y alone, so the two points are $(-\cos \theta, \sin \theta)$ and $(\cos \theta, \sin \theta)$. Comparing coordinates:

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta.$$

Reflect in the x -axis. Measuring $-\theta$ means turning clockwise instead of anticlockwise, which lands on the mirror image of P across the x -axis. That reflection negates y and leaves x alone, so

$$\cos(-\theta) = \cos \theta, \quad \sin(-\theta) = -\sin \theta.$$

Turn a full circle. Adding 2π to the angle sends P once round and back to exactly the same point, so both coordinates are unchanged:

$$\sin(\theta + 2\pi) = \sin \theta, \quad \cos(\theta + 2\pi) = \cos \theta.$$

This is why both functions are **periodic** with period 2π . Every identity of this kind says the same thing: move the point, see which coordinate changes sign.

You have met the first of these relations before. Two angles, θ and $\pi - \theta$, share the same sine. That shared value is what produced the ambiguous case of the sine rule in Ch. 7. A triangle built from a sine value can sometimes close in two ways, one acute and one obtuse.

An exact value at any special angle now comes out of the circle in two moves: find the acute angle the arm makes with the x -axis, which fixes the size, then look at the quadrant, which fixes the sign.

EXAMPLE 8.1

Find the exact values of $\sin \frac{5\pi}{3}$ and $\cos \frac{5\pi}{3}$.

Write the angle as $2\pi - \frac{\pi}{3}$. A full turn returns to the positive x -axis, so turning back through $\frac{\pi}{3}$ leaves the point in the fourth quadrant, where the x -coordinate is positive and the y -coordinate negative.

The arm makes an acute angle of $\frac{\pi}{3}$ with the axis, so both values have the size they have at $\frac{\pi}{3}$:

$$\sin \frac{5\pi}{3} = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}, \quad \cos \frac{5\pi}{3} = \cos \frac{\pi}{3} = \frac{1}{2}.$$

Dividing them gives $\tan \frac{5\pi}{3} = -\sqrt{3}$, negative, because only the sine carries a minus sign. That is the fourth quadrant's entry in the mnemonic, recovered from the coordinates rather than remembered.

YOUR TURN 8.1 The Unit Circle

1. Write down the exact value of each.

(a) $\sin \frac{\pi}{6}$
(c) $\tan \frac{\pi}{3}$

(b) $\cos \frac{\pi}{4}$
(d) $\tan \frac{\pi}{4}$

2. Use the symmetry of the circle to find each exact value.

(a) $\sin \frac{5\pi}{6}$
(c) $\sin \frac{4\pi}{3}$

(b) $\cos \frac{2\pi}{3}$
(d) $\cos \frac{7\pi}{4}$

3. State the quadrant or quadrants in which each condition holds.

(a) $\sin \theta > 0$
(c) $\tan \theta > 0$

(b) $\cos \theta < 0$
(d) $\sin \theta < 0$ and $\cos \theta > 0$

4. The point P on the unit circle has angle θ .

- (a) State the coordinates of P in terms of θ .
(b) Explain why $\cos^2 \theta + \sin^2 \theta = 1$ for every θ .
(c) Explain why $\tan \theta$ is undefined at $\theta = \frac{\pi}{2}$.

5. Lines through the origin, using $y = x \tan \theta$. Give each angle in radians.

- (a) Write the equation of the line through the origin making an angle of $\frac{\pi}{3}$ with the positive x -axis.
(b) Find the angle a line of gradient 1 makes with the positive x -axis.
(c) Find the anticlockwise angle from the positive x -axis to the line $y = -x$.

8.2 Trigonometric Identities

An **identity** is an equation true for every angle. The most fundamental one is Pythagoras' theorem read off the unit circle: the point $(\cos \theta, \sin \theta)$ is distance 1 from the origin, so $\cos^2 \theta + \sin^2 \theta = 1$.

KEY · IB $\cos^2 \theta + \sin^2 \theta = 1$.

8.2.1 Double and compound angles

The functions of a sum of two angles can be expanded, and from the sum formulas everything else follows. These **compound-angle** identities are usually just quoted, but the proof is short and uses only the unit circle and the cosine rule from Ch. 7.

Put two points on the unit circle, P at angle A and Q at angle B . Their coordinates are $P(\cos A, \sin A)$ and $Q(\cos B, \sin B)$, and the angle $\angle POQ$ at the centre is $A - B$.

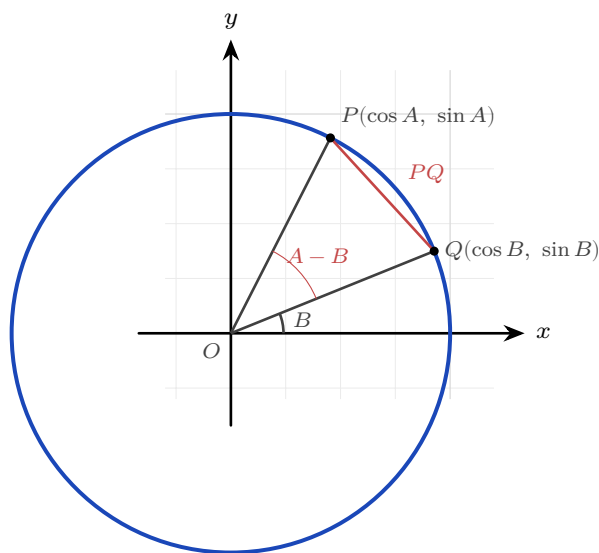


Figure 8.3 Two points on the unit circle, at angles A and B . The angle between the radii is $A - B$, and the chord PQ can be measured either by the distance formula or by the cosine rule.

Now find the distance PQ in two different ways.

By the distance formula:

$$PQ^2 = (\cos A - \cos B)^2 + (\sin A - \sin B)^2.$$

Expanding gives $\cos^2 A - 2 \cos A \cos B + \cos^2 B + \sin^2 A - 2 \sin A \sin B + \sin^2 B$. Both Pythagorean pairs collapse to 1, so

$$PQ^2 = 2 - 2(\cos A \cos B + \sin A \sin B).$$

By the cosine rule in triangle OPQ , where both sides from the centre have length 1 and the angle between them is $A - B$:

$$PQ^2 = 1^2 + 1^2 - 2(1)(1) \cos(A - B) = 2 - 2 \cos(A - B).$$

The two expressions are the same length, so

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Every other compound-angle formula follows from this one. Replacing B by $-B$, and using

$\cos(-B) = \cos B$ with $\sin(-B) = -\sin B$, gives the formula for $\cos(A + B)$.

A sine is a cosine a quarter turn away: $\sin \theta = \cos(\frac{\pi}{2} - \theta)$. That single fact is all the sine version needs. Write the sum in that form, then group the result as a *difference* of two angles:

$$\sin(A + B) = \cos(\frac{\pi}{2} - A - B) = \cos((\frac{\pi}{2} - A) - B).$$

That is exactly the shape just proved. Expanding it, and using $\cos(\frac{\pi}{2} - A) = \sin A$ together with $\sin(\frac{\pi}{2} - A) = \cos A$, gives

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

The tangent versions come from dividing the sine formula by the cosine formula.

KEY · IB HL Compound angles:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B, \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

Setting $B = A$ reduces each of these to a **double-angle** identity, the most useful family in the topic. Working through the derivation once shows there is nothing new to learn here.

Put $B = A$ in the sine formula:

$$\sin 2A = \sin(A + A) = \sin A \cos A + \cos A \sin A = 2 \sin A \cos A.$$

Do the same in the cosine formula:

$$\cos 2A = \cos(A + A) = \cos A \cos A - \sin A \sin A = \cos^2 A - \sin^2 A.$$

That is the first form. The other two come from $\sin^2 A + \cos^2 A = 1$, used to remove one function or the other. Replacing $\sin^2 A$ by $1 - \cos^2 A$ gives

$$\cos 2A = \cos^2 A - (1 - \cos^2 A) = 2 \cos^2 A - 1,$$

and replacing $\cos^2 A$ by $1 - \sin^2 A$ instead gives

$$\cos 2A = (1 - \sin^2 A) - \sin^2 A = 1 - 2 \sin^2 A.$$

The tangent version follows in the same way, by putting $B = A$ in the tangent formula.

The sine and cosine versions sit in the SL half of the booklet; the tangent version is HL.

KEY · IB Double angles:

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

$$\text{KEY · IB HL} \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

The three forms of $\cos 2\theta$ are one identity rewritten by $\sin^2 + \cos^2 = 1$. Choose whichever leaves you with only sines or only cosines. That choice is what makes the identity so useful when solving equations.

EXAMPLE 8.2

Find the exact value of $\sin 165^\circ$.

Write $165^\circ = 120^\circ + 45^\circ$ and use the compound-angle formula:

$$\sin 165^\circ = \sin 120^\circ \cos 45^\circ + \cos 120^\circ \sin 45^\circ = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right) \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}.$$

Splitting an awkward angle into a sum or a difference of two known ones is the standard method for exact values beyond the basic table. The pieces need not be acute: $\cos 120^\circ = -\frac{1}{2}$ carries a minus sign from the second quadrant, and that minus sign is what turns the sum into a difference.

Identities also let you move between ratios *without ever finding the angle*. Examiners set this often: the angle stays unknown from start to finish, and the Pythagorean identity supplies everything needed.

EXAMPLE 8.3

Given that $\cos x = \frac{3}{4}$ and x is acute, find the exact value of $\sin 2x$ without finding x .

From $\cos^2 x + \sin^2 x = 1$, $\sin^2 x = 1 - \frac{9}{16} = \frac{7}{16}$, and since x is acute, $\sin x = \frac{\sqrt{7}}{4}$ (positive).

Then

$$\sin 2x = 2 \sin x \cos x = 2 \cdot \frac{\sqrt{7}}{4} \cdot \frac{3}{4} = \frac{3\sqrt{7}}{8}.$$

The word “acute” fixed the sign of $\sin x$. If x had instead been in the fourth quadrant, the same working would give $\sin 2x = -\frac{3\sqrt{7}}{8}$: always let the quadrant choose the sign before doubling.

Proving an identity is the same game played in reverse: start from one side, substitute known identities, and simplify until the other side appears.

EXAMPLE 8.4

Prove that $\cos(A - B) - \cos(A + B) = 2 \sin A \sin B$.

Work on the left side, expanding each cosine by the compound-angle formula:

$$\cos(A - B) = \cos A \cos B + \sin A \sin B, \quad \cos(A + B) = \cos A \cos B - \sin A \sin B.$$

The two expansions differ only in the sign of the second term, so subtracting removes the $\cos A \cos B$ pair and doubles what is left:

$$\cos(A - B) - \cos(A + B) = 2 \sin A \sin B. \quad \blacksquare$$

Expanding the more complicated side first is the usual choice. Once both expansions are written down, the terms that have to cancel are visible before any algebra is done.

8.2.2 One wave from two

A sum such as $8 \sin x + 15 \cos x$ mixes a sine and a cosine of the same angle. Both have period 2π , so the sum repeats every 2π as well, and it turns out to be a single sine wave, stretched and shifted. Reading the compound-angle formula backwards shows why.

Expanding $R \sin(x + \alpha)$ gives

$$R \sin(x + \alpha) = R \cos \alpha \sin x + R \sin \alpha \cos x,$$

which already has the shape $a \sin x + b \cos x$, with $a = R \cos \alpha$ and $b = R \sin \alpha$. Squaring those two and adding removes α , since $\cos^2 \alpha + \sin^2 \alpha = 1$, and leaves $a^2 + b^2 = R^2$. Dividing them removes R instead and leaves $\tan \alpha = \frac{b}{a}$. So any such sum can be written as one wave, and the two constants can be recovered from a and b alone.

KEY **The wave form.** For constants a and b not both zero,

$$a \sin x + b \cos x = R \sin(x + \alpha), \quad R = \sqrt{a^2 + b^2}, \quad \tan \alpha = \frac{b}{a},$$

with $R > 0$ and α taken in the quadrant where $\cos \alpha$ has the sign of a and $\sin \alpha$ the sign of b . The same R and α give the cosine form

$$a \cos x + b \sin x = R \cos(x - \alpha).$$

Since $\sin$ and $\cos$ run between -1 and 1 , the expression has maximum R and minimum $-R$.

This form is not printed in the formula booklet, so rebuild it from the expansion of $R \sin(x + \alpha)$ whenever it is needed. Its value is that the variable now appears once instead of twice, which turns a maximum, a minimum or an equation into the ordinary work of §8.6.

EXAMPLE 8.5

Write $8 \sin x + 15 \cos x$ in the form $R \sin(x + \alpha)$ with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. State the maximum and minimum values, and solve $8 \sin x + 15 \cos x = 6$ for $0 \leq x \leq 2\pi$.

Here $a = 8$ and $b = 15$. Both are positive, so α is in the first quadrant:

$$R = \sqrt{8^2 + 15^2} = \sqrt{289} = 17, \quad \tan \alpha = \frac{15}{8} \implies \alpha = \arctan \frac{15}{8} = 1.0808,$$

giving $8 \sin x + 15 \cos x = 17 \sin(x + 1.0808)$.

The sine factor runs between -1 and 1 , so the maximum is 17 and the minimum is -17 .

For the equation, $17 \sin(x + \alpha) = 6$, so $\sin(x + \alpha) = \frac{6}{17}$. Put $u = x + \alpha$. As x runs over $[0, 2\pi]$, u runs over $[1.0808, 7.3640]$. The reference angle is $\arcsin \frac{6}{17} = 0.3607$, and $\frac{6}{17}$ is positive, so sine takes it in the first and second quadrants:

$$u = 0.3607 \quad \text{and} \quad u = \pi - 0.3607 = 2.7809.$$

The first sits below 1.0808 , so add a turn: $0.3607 + 2\pi = 6.6439$, which is inside. The second is already inside, and a further turn from it would pass 7.3640 . So $u = 2.7809$ or $u = 6.6439$, and subtracting α ,

$$x = 1.70 \quad \text{or} \quad x = 5.56 \quad (3 \text{ s.f.}).$$

Transforming the interval along with the substitution is what saved the second answer. Had u been left to run over $[0, 2\pi]$, the pair 0.3607 and 2.7809 would have given one negative x and lost 5.56 altogether.

YOUR TURN 8.2 Trigonometric Identities

6. Simplify each expression.

(a) $\sin^2 \theta + \cos^2 \theta$

(b) $2 \sin \theta \cos \theta$

(c) $\cos^2 \theta - \sin^2 \theta$

(d) $1 - 2 \sin^2 \theta$

7. Given $\sin \theta = \frac{3}{5}$ with θ acute, find each exact value.

(a) $\cos \theta$

(b) $\tan \theta$

(c) $\sin 2\theta$

(d) $\cos 2\theta$

8. Given $\cos \theta = -\frac{3}{5}$ with θ in the second quadrant, find each exact value.

(a) $\sin \theta$

(b) $\tan \theta$

(c) $\sin 2\theta$

(d) $\cos 2\theta$

9. Use a compound-angle formula to find each exact value.

(a) $\cos 15^\circ$, from $45^\circ - 30^\circ$

(b) $\sin 75^\circ$, from $45^\circ + 30^\circ$

10. Prove each identity.

(a) $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$

(b) $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$

(c) $(\sin \theta + \cos \theta)^2 = 1 + \sin 2\theta$

(d) $\frac{\cos 2\theta}{\cos \theta + \sin \theta} = \cos \theta - \sin \theta$

(e) $\tan \theta \sin 2\theta = 2 \sin^2 \theta$

(f) $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$

8.3 Graphs of Trigonometric Functions

The graphs are the circle unrolled. Send the point P round the circle and plot its y -coordinate against the angle: that plot is $y = \sin x$. Plot its x -coordinate instead and you get $y = \cos x$.

Every feature of the sine graph can be read off the circle before you plot a single point.

- The y -coordinate of P never leaves -1 to 1 , so the curve stays between those two heights. Its **amplitude**, the distance from the middle to the top, is 1 .
- P returns to its starting place after one full turn, so the curve repeats every 2π . Its **period**, the width of one complete cycle, is 2π .
- $y = 0$ when P sits on the x -axis, at $\theta = 0, \pi$ and 2π , so the curve crosses the axis at those three angles.
- y is largest at the top of the circle, at $\theta = \frac{\pi}{2}$, so the maximum is 1 and it occurs at $x = \frac{\pi}{2}$.
- $y > 0$ everywhere in the upper half of the circle, so the curve is above the axis for $0 < x < \pi$.

Read the x -coordinate instead and the cosine graph appears the same way. It starts at 1 , because P starts on the positive x -axis, and falls to 0 at $\frac{\pi}{2}$, where P is at the top and its x -coordinate is zero.

So both curves have amplitude 1 and period 2π . They differ only in where a cycle begins. The x -coordinate at angle θ is the y -coordinate a quarter turn later, so $\cos \theta = \sin(\theta + \frac{\pi}{2})$, and the cosine curve is the sine curve shifted left by $\frac{\pi}{2}$.

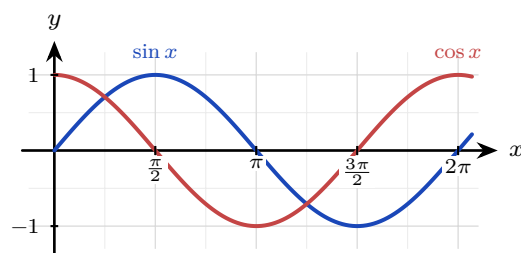


Figure 8.4 One turn of the circle unrolled. Both waves have amplitude 1 and period 2π ; the cosine is the sine shifted left by $\frac{\pi}{2}$.

The tangent behaves differently, and the circle explains why. Since $\tan \theta = \frac{y}{x}$, it is undefined wherever $x = 0$, which is at the top and the bottom of the circle. That gives vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.

Its period is π , not 2π . Half a turn sends P to the point diametrically opposite, whose coordinates are $(-x, -y)$. The ratio $\frac{-y}{-x}$ equals $\frac{y}{x}$, so the tangent repeats after only half a revolution.

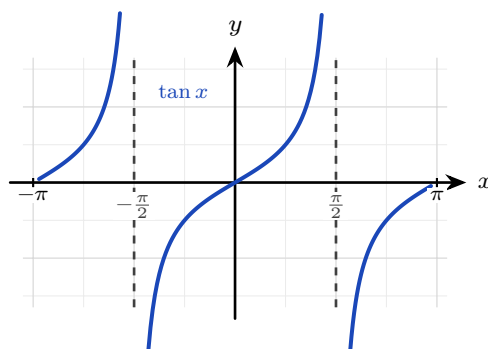


Figure 8.5 The tangent repeats every π , not every 2π , with a vertical asymptote wherever $\cos x = 0$.

That is the basic behaviour, collected in one place.

	$\sin x$	$\cos x$	$\tan x$
Domain	all x	all x	$x \neq \frac{\pi}{2} + n\pi$
Range	$-1 \leq y \leq 1$	$-1 \leq y \leq 1$	all y
Period	2π	2π	π
Amplitude	1	1	none
Zeros	$n\pi$	$\frac{\pi}{2} + n\pi$	$n\pi$

Here n stands for any integer, and the last row gives the values of x at which the curve crosses the axis. The tangent has no amplitude because it has no largest or smallest value. Amplitude measures half the rise and fall, and the tangent rises without limit.

8.3.1 Transforming the wave

The general sinusoid stretches and slides the basic sine, and each parameter has a plain meaning.

KEY For $y = a \sin(b(x + c)) + d$:

$$\begin{aligned} \text{amplitude} &= |a|, & \text{period} &= \frac{2\pi}{|b|}, \\ \text{horizontal shift} &= -c, & \text{vertical shift (midline)} &= d. \end{aligned}$$

Questions often run the other way, giving you the graph's largest and smallest values and asking for a and d . Since the maximum is $d + |a|$ and the minimum is $d - |a|$, adding and subtracting recovers both constants.

- **Amplitude.** From the equation it is $|a|$. From a graph it is $\frac{\max - \min}{2}$, half the total rise and fall.
- **Midline.** From the equation it is the line $y = d$. From a graph it is $y = \frac{\max + \min}{2}$, the level halfway between the top and the bottom.
- **Maximum and minimum.** They are $d + |a|$ and $d - |a|$: the midline, plus and minus the amplitude.
- **Period.** From the equation it is $\frac{2\pi}{|b|}$. From a graph it is the horizontal distance from one repeat to the next.

Read the first two as sentences and they stop being formulas to memorise. Consider the wave $y = 5 \sin(2x) - 3$, whose maximum is 2 and whose minimum is -8 . The amplitude is $\frac{2 - (-8)}{2} = 5$ and the midline is $\frac{2 + (-8)}{2} = -3$, which is what the equation says.

EXAMPLE 8.6

Describe $y = 2 \sin(3x) + 1$ and state its range.

The amplitude is 2, so the wave rises and falls 2 either side of its midline. The period is $\frac{2\pi}{3}$, so it completes a full cycle three times as fast as $\sin x$. The midline is $y = 1$. The graph therefore runs from $1 - 2 = -1$ up to $1 + 2 = 3$, giving range $-1 \leq y \leq 3$.

8.3.2 Symmetry of the graphs HL

Each reflection symmetry of the circle from Section 8.1 becomes a visible symmetry of the graphs. The cosine curve is mirror-symmetric in the y -axis, so $\cos(-x) = \cos x$: cosine is an **even** function. The sine and tangent curves map onto themselves under a half-turn about the origin, so $\sin(-x) = -\sin x$ and $\tan(-x) = -\tan x$: both are **odd**, in the sense of Ch. 5. The same circle symmetries give the supplementary-angle relations.

KEY **HL** $\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta, \quad \tan(\pi - \theta) = -\tan \theta.$

You can see all three on the graphs directly: the sine curve is symmetric about the vertical line $x = \frac{\pi}{2}$, while cosine and tangent flip sign across it. In an exam these relations remove the need for a calculator: $\cos \frac{3\pi}{4} = -\cos \frac{\pi}{4} = -\frac{\sqrt{2}}{2}$.

YOUR TURN 8.3 Graphs of Trigonometric Functions

11. State the amplitude, the period and the midline of each graph.

(a) $y = 3 \sin x$

(b) $y = \sin 2x$

(c) $y = 4 \sin(3x) - 2$

(d) $y = 5 \sin(4x) + 3$

12. State the range of each function.

(a) $y = 2 \sin x + 1$

(b) $y = 3 \sin(2x) - 1$

(c) $y = 2 \cos\left(\frac{x}{2}\right) - 1$

13. Answer each question about $y = \tan x$.

(a) State its period.

(b) State the equations of two of its asymptotes.

(c) Explain why the asymptotes occur where they do.

14. A wheel of diameter 40 m has its lowest point 2 m above the ground and turns once every 20 seconds. A seat starts at the bottom.

(a) State the amplitude and the midline of the seat's height.

(b) Find the value of b in $h(t) = d - a \cos(bt)$.

(c) Write down $h(t)$.

(d) Find the height after 5 seconds.

8.4 Reciprocal Trigonometric Functions **HL**

Divide 1 by a sine, a cosine or a tangent and three more functions appear. Each has a name of its own, a graph with vertical asymptotes, and an identity that makes certain equations solvable.

KEY · IB $\sec \theta = \frac{1}{\cos \theta}, \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta}.$

The third is the **cotangent**, $\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$. Its definition is not printed in the formula booklet, so learn it.

The pairing looks wrong at first: secant goes with cosine, and cosecant with sine. There is no reason behind it, so use the third letter. In **sec** the third letter is c, for cosine. In **cosec** it is s, for sine. In **cot** it is t, for tangent.

Two facts follow from the reciprocal alone, and between them they fix every graph.

First, a reciprocal has an asymptote wherever the original function is zero, because you cannot divide by zero. So sec has asymptotes where $\cos \theta = 0$, at $\theta = \frac{\pi}{2} + n\pi$, while cosec and cot have them where $\sin \theta = 0$, at $\theta = n\pi$.

Second, sin and cos never exceed 1 in size, so their reciprocals are never smaller than 1 in size. Neither $\sec \theta$ nor $\operatorname{cosec} \theta$ ever takes a value strictly between -1 and 1 . That gap in the middle is the feature examiners test most often.

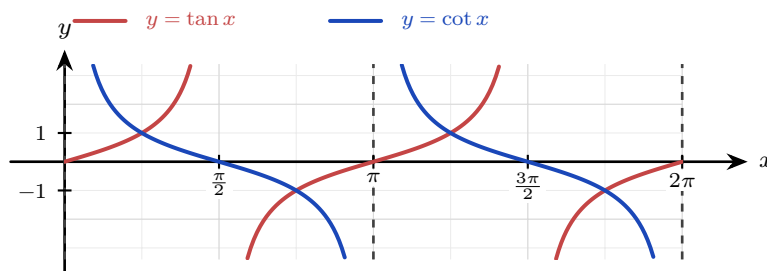
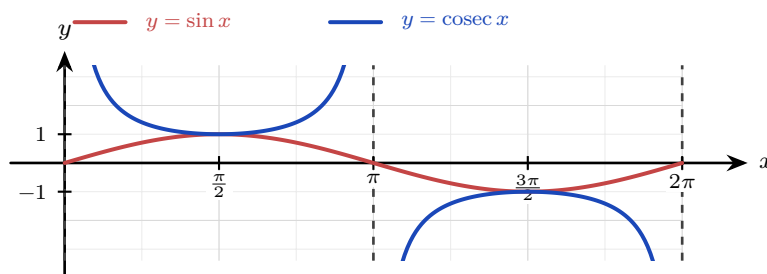
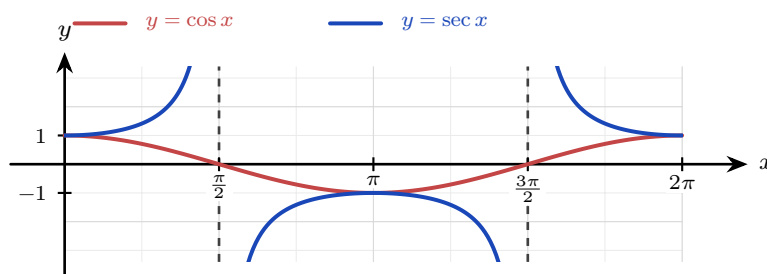


Figure 8.6 In each panel the parent function is scarlet and its reciprocal cobalt. The reciprocal has an asymptote, marked by a dashed line, wherever its parent crosses the axis, and the two curves meet wherever the parent reaches ± 1 . Secant and cosecant never enter the band between -1 and 1 ; cotangent has no such gap.

Taking n to be any integer, the three read as follows.

Function	Domain	Range	Period
$\sec \theta = \frac{1}{\cos \theta}$	all θ except $\frac{\pi}{2} + n\pi$	$y \leq -1$ or $y \geq 1$	2π
$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	all θ except $n\pi$	$y \leq -1$ or $y \geq 1$	2π
$\cot \theta = \frac{\cos \theta}{\sin \theta}$	all θ except $n\pi$	all real y	π

Every excluded value in the domain column is a zero of the function underneath, and every one of them is an asymptote on the graph.

8.4.1 The two Pythagorean variants

The identity $\cos^2 \theta + \sin^2 \theta = 1$ has two companions, one for each reciprocal pair, and both come from dividing it through.

Dividing $\cos^2 \theta + \sin^2 \theta = 1$ by $\cos^2 \theta$ gives

$$1 + \tan^2 \theta = \sec^2 \theta,$$

since $\frac{\sin \theta}{\cos \theta} = \tan \theta$ and $\frac{1}{\cos \theta} = \sec \theta$. Dividing the same identity by $\sin^2 \theta$ instead gives the companion result. Deriving them when you need them is safer than trusting your memory of which is which.

KEY · IB $1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta.$

These identities exist to turn mixed equations into single-function ones, exactly as the double angles did.

EXAMPLE 8.7

Solve $\tan^2 x + \sec x = 1$ for $0 \leq x \leq 2\pi$.

The equation mixes $\tan$ and $\sec$; the identity $\tan^2 x = \sec^2 x - 1$ rewrites it in $\sec x$ alone:

$$\sec^2 x - 1 + \sec x = 1 \implies \sec^2 x + \sec x - 2 = 0 \implies (\sec x + 2)(\sec x - 1) = 0.$$

So $\sec x = -2$ or $\sec x = 1$, which is $\cos x = -\frac{1}{2}$ or $\cos x = 1$. From the circle,

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{or} \quad x = 0, 2\pi.$$

Converting back to $\cos$ at the end is the safest final step, because the unit circle is read in sine and cosine, not in secant.

YOUR TURN 8.4 Reciprocal Trigonometric Functions HL

15. Evaluate each exactly.

(a) $\sec \frac{\pi}{3}$

(c) $\cot \frac{\pi}{4}$

(e) $\cot \frac{2\pi}{3}$

(b) $\operatorname{cosec} \frac{\pi}{2}$

(d) $\sec \frac{2\pi}{3}$

(f) $\operatorname{cosec} \frac{7\pi}{6}$

16. Simplify each expression.

(a) $\sec^2 \theta - \tan^2 \theta$

(c) $\frac{1}{\cot \theta}$

(e) $\frac{\sec \theta}{\operatorname{cosec} \theta}$

(b) $\operatorname{cosec}^2 \theta - \cot^2 \theta$

(d) $\tan \theta \cot \theta$

(f) $\sin \theta \operatorname{cosec} \theta + \cos \theta \sec \theta$

17. Prove each identity.

(a) $\cot \theta \sec \theta = \operatorname{cosec} \theta$

(b) $\frac{\operatorname{cosec} \theta}{\cot \theta + \tan \theta} = \cos \theta$

(c) $\sec^2 \theta + \operatorname{cosec}^2 \theta = \sec^2 \theta \operatorname{cosec}^2 \theta$

(d) $\frac{1 + \cot^2 \theta}{1 + \tan^2 \theta} = \cot^2 \theta$

8.5 Inverse Trigonometric Functions HL

Everything so far has gone one way: give an angle, get a ratio. The inverse functions run the other way, recovering the angle from the ratio.

You have already used them without naming them. When you solve $\sin x = \frac{1}{2}$ and press the $\sin^{-1}$ key, that key is the inverse function.

Two notations, one function. The inverse of sine is written either $\arcsin x$ or $\sin^{-1} x$.

They mean exactly the same thing. These notes use $\arcsin$, because the other form invites a serious confusion:

$$\sin^{-1} x \text{ means the inverse, but } (\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x.$$

The -1 is not an exponent here. Writing $\arcsin$, $\arccos$ and $\arctan$ removes the ambiguity, and your calculator's $\sin^{-1}$ key is the same function under the other name.

8.5.1 Why a restriction is needed

There is an immediate difficulty. A function has an inverse only if it is one-to-one (Ch. 5), and $\sin$, $\cos$ and $\tan$ are as far from one-to-one as a function can be. Each takes every value in its range infinitely often, so the horizontal line $y = \frac{1}{2}$ meets $y = \sin x$ infinitely many times.

That is a real problem, not a technicality. If you ask “which angle has sine $\frac{1}{2}$?”, there is no single answer: $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{13\pi}{6}$ and infinitely many others all qualify. A function must return one output, so it has to choose.

The fix is the one from Ch. 5: cut the domain down until the function is one-to-one, then invert what is left. Which piece to keep is a choice, and mathematicians have agreed on one for each function. Those agreed pieces are shaded in the diagrams below, and they are what fix the ranges in the table.

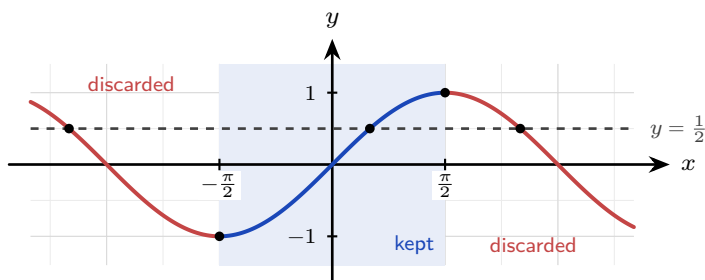


Figure 8.7 The dashed line $y = \frac{1}{2}$ meets $y = \sin x$ again and again, which is why “the angle whose sine is $\frac{1}{2}$ ” has no single answer. Keeping only the shaded piece from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ leaves one intersection. On that piece the curve rises steadily, so it is one-to-one and every output comes from exactly one input.

The same is done for the other two: cosine keeps $0 \leq x \leq \pi$, where it falls steadily, and tangent keeps $-\frac{\pi}{2} < x < \frac{\pi}{2}$, one branch between two asymptotes. Those kept pieces become the *ranges* of the inverses, because domain and range swap when a function is inverted.

Function	Domain	Range
$\arcsin x$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
$\arccos x$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
$\arctan x$	all real x	$-\frac{\pi}{2} < y < \frac{\pi}{2}$

Each inverse is a reflection of its restricted parent in the line $y = x$ (Ch. 5), so the graphs are the familiar curves turned on their sides. Arcsine climbs steeply from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$. Arccosine falls from $(-1, \pi)$ to $(1, 0)$. Arctangent flattens towards the horizontal asymptotes $y = \pm\frac{\pi}{2}$, which are the vertical asymptotes of tangent after the reflection.

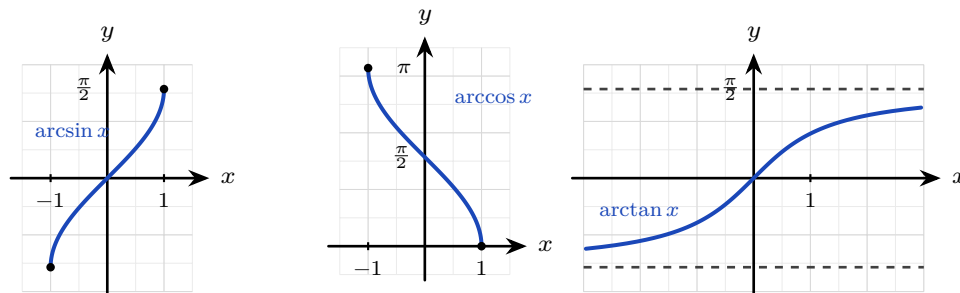


Figure 8.8 Each inverse is its restricted parent reflected in $y = x$. The marked endpoints show the range each one is confined to.

EXAMPLE 8.8

Find the exact values of (a) $\arcsin(-\frac{\sqrt{3}}{2})$, (b) $\arccos(-\frac{\sqrt{3}}{2})$, (c) $\arcsin(\sin \frac{5\pi}{4})$.

(a) The angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is $-\frac{\sqrt{3}}{2}$ is $-\frac{\pi}{3}$. Not $\frac{4\pi}{3}$ and not $\frac{5\pi}{3}$: those solve the equation but lie outside arcsine's range.

(b) The angle in $[0, \pi]$ whose cosine is $-\frac{\sqrt{3}}{2}$ is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Arccosine handles negatives by reaching into the second quadrant.

(c) $\sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$, and $\arcsin(-\frac{\sqrt{2}}{2}) = -\frac{\pi}{4}$. The composition returns $-\frac{\pi}{4}$, not $\frac{5\pi}{4}$: arcsine can only answer with angles from its own range.

CAUTION $\arcsin(\sin x)$ does not always give back x . Every answer from arcsin must lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, and 2 does not. So $\arcsin(\sin 2) = \pi - 2$, an angle with the same sine as 2 but inside the range. The restriction is working, not failing.

8.5.2 Compositions the other way round

A composition can also run with the inverse on the inside, as in $\cos(\arcsin k)$. The angle is never found here, and it never needs to be. Name it: let $\theta = \arcsin k$, so $\sin \theta = k$. A right triangle with opposite k and hypotenuse 1 has exactly that sine, and Pythagoras gives its adjacent side as $\sqrt{1 - k^2}$, so

$$\cos(\arcsin k) = \sqrt{1 - k^2}.$$

The positive root is the right one because arcsin answers in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, and cosine is never negative there. The same triangle serves arccos and arctan: place the known ratio on the two sides it names, find the third by Pythagoras, then read off whatever the question asks for.

EXAMPLE 8.9

Find the exact value of (a) $\cos\left(\arcsin \frac{20}{29}\right)$, (b) $\sin\left(\arccos \frac{9}{41} + \arctan \frac{20}{21}\right)$.

(a) Let $\theta = \arcsin \frac{20}{29}$, so $\sin \theta = \frac{20}{29}$: opposite 20, hypotenuse 29. The adjacent side is $\sqrt{29^2 - 20^2} = \sqrt{441} = 21$, so

$$\cos\left(\arcsin \frac{20}{29}\right) = \frac{21}{29},$$

positive because θ lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

(b) Name both angles: $A = \arccos \frac{9}{41}$ and $B = \arctan \frac{20}{21}$. A 9–40–41 triangle gives $\cos A = \frac{9}{41}$ and $\sin A = \frac{40}{41}$, taken positive because $\arccos$ answers in $[0, \pi]$, where sine is never negative. A 20–21–29 triangle, the same one as in part (a), gives $\sin B = \frac{20}{29}$ and $\cos B = \frac{21}{29}$, both positive because $\tan B > 0$ puts B in the first quadrant. The sum is now a compound angle:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B = \frac{40}{41} \cdot \frac{21}{29} + \frac{9}{41} \cdot \frac{20}{29} = \frac{840 + 180}{1189} = \frac{1020}{1189}.$$

Neither A nor B was ever evaluated. The range of each inverse function is what fixes the sign of every side, and that is the only place in the working where a decision is made.

YOUR TURN 8.5 Inverse Trigonometric Functions HL

18. Evaluate each exactly.

(a) $\arcsin \frac{1}{2}$

(b) $\arccos 0$

(c) $\arctan(-1)$

(d) $\arccos\left(-\frac{1}{2}\right)$

19. An inverse undoes its function only on the piece that was kept.

(a) State the range of $\arcsin x$, of $\arccos x$ and of $\arctan x$.

(b) Evaluate $\arcsin\left(\sin \frac{3\pi}{4}\right)$, and explain why the answer is not $\frac{3\pi}{4}$.

(c) Evaluate $\arccos\left(\cos \frac{4\pi}{3}\right)$.

(d) Evaluate $\arctan\left(\tan \frac{5\pi}{6}\right)$.

(e) State every value of x for which $\arcsin(\sin x) = x$.

20. GDC Find the exact value, or write **undefined**. Check each on your GDC.

(a) $\arcsin\left(\sin \frac{2\pi}{3}\right)$

(b) $\arccos\left(\cos \frac{3}{2}\right)$

(c) $\tan(\arctan 12)$

(d) $\cos\left(\arccos \frac{2\pi}{3}\right)$

(e) $\arctan\left(\tan\left(-\frac{3\pi}{4}\right)\right)$

(f) $\sin(\arcsin \pi)$

(g) $\sin\left(\arctan \frac{3}{4}\right)$

(i) $\arcsin\left(\tan \frac{\pi}{3}\right)$

(k) $\cos\left(\arctan \frac{1}{2}\right)$

(m) $\sin\left(\arccos \frac{3}{5} + \arctan \frac{5}{12}\right)$

(h) $\cos\left(\arcsin \frac{7}{25}\right)$

(j) $\arctan\left(2 \sin \frac{\pi}{3}\right)$

(l) $\cos(\arcsin 0.6)$

(n) $\cos\left(\arctan 3 + \arcsin \frac{1}{3}\right)$

8.6 Solving Trigonometric Equations

An equation like $\sin x = \frac{1}{2}$ has not one solution but infinitely many, because the sine repeats every 2π . Your calculator returns only one of them, the **principal value**. Everything else you must find yourself, and the circle is what tells you how.

Start by counting the solutions in a single turn. Solving $\sin x = k$ means finding the points on the circle whose y -coordinate is k , so draw the horizontal line at height k and see where it meets the circle. Three things can happen.

- If $-1 < k < 1$ the line cuts the circle in **two** places, so two angles in each turn have sine k .
- If $k = 1$ or $k = -1$ the line touches the circle at **one** point, the top or the bottom. The two angles have merged into one.
- If $|k| > 1$ the line misses the circle, and there is **no** solution at all.

The same three cases hold for $\cos x = k$, read with a vertical line at $x = k$ instead. Tangent behaves differently: it takes every real value, so $\tan x = k$ has solutions whatever k is.

Two words are needed next. You will use both constantly.

DEF The **reference angle** $\hat{\theta}$ is the acute angle between the arm OP and the x -axis. It is positive, it is less than $\frac{\pi}{2}$, and it ignores signs entirely: you get it from the size of k alone, as $\arcsin |k|$, $\arccos |k|$ or $\arctan |k|$. It is defined whenever the arm lies off the axes.

The **principal value** is the single angle your calculator returns. It is the one solution lying inside the range of the matching inverse function:

$$\arcsin k \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \arccos k \in [0, \pi], \quad \arctan k \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

The two are easy to confuse. A principal value can be negative, and for a cosine it can be obtuse; a reference angle is never either. Take $\cos x = -\frac{1}{2}$. The calculator returns the principal value $\frac{2\pi}{3}$, while the reference angle is $\frac{\pi}{3}$, the acute angle the arm makes with the neg-

ative x -axis. Work from the reference angle and the calculator's choice of quadrant stops mattering.

The method. Every solution is built from the reference angle and the quadrant it sits in. The sign of k tells you which two quadrants to use, by the rule of §8.1.2, and each quadrant has one construction.

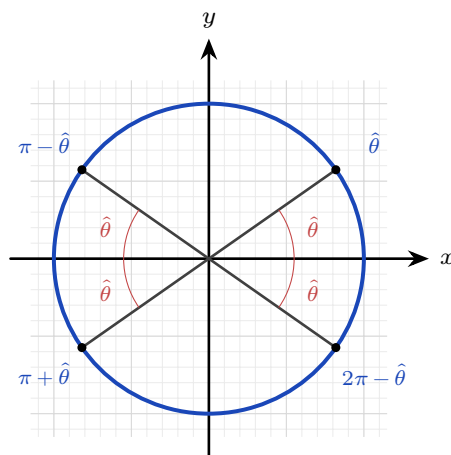


Figure 8.9 One reference angle $\hat{\theta}$, measured from the x -axis, gives an angle in each quadrant. The scarlet arcs are all the same size; only the direction they are measured from changes.

KEY Solving a trigonometric equation.

1. Rearrange until a single trigonometric function is alone on one side.
2. If the argument is not plain x , substitute u for it and **transform the interval as well**.
3. Find the reference angle from the *size* of k : $\hat{\theta} = \arcsin |k|$, $\arccos |k|$ or $\arctan |k|$.
4. Use the sign of k to choose the two quadrants, then build the angle in each: $\hat{\theta}$, $\pi - \hat{\theta}$, $\pi + \hat{\theta}$ or $2\pi - \hat{\theta}$.
5. Add or subtract whole turns of 2π until you have every solution the interval asks for.
6. If you substituted, convert back to x at the very end.

There is a shorter route once you already have one solution, which you always do: the calculator has just given you one. Call it θ . Its partner in the same turn is fixed by the function alone, whatever quadrant θ landed in.

KEY The partner solution. If θ is one solution, then within $0 \leq x \leq 2\pi$ the other is

$$\sin x = k : \quad \pi - \theta, \quad \cos x = k : \quad 2\pi - \theta, \quad \tan x = k : \quad \pi + \theta.$$

Then add or subtract whole turns of 2π — for the tangent, half turns of π — to reach every solution the interval asks for.

The interval decides how the partner is written, not what it is. On $-\pi \leq x \leq \pi$ the cosine

pair is θ and $-\theta$, because $2\pi - \theta$ is one whole turn away from $-\theta$. That is why a cosine equation on a symmetric interval always answers $\pm$ the same number, and it is worth recognising on sight: $\cos x = \frac{1}{2}$ on $-\pi \leq x \leq \pi$ gives $x = \pm \frac{\pi}{3}$ and nothing else.

Both routes give the same angles. The reference angle is the one to fall back on when the signs get confusing, or when the equation has been rearranged so far that you no longer trust which quadrant you are in.

Two checks belong in every solution. Your two angles should lie in the quadrants where that function has the sign of k ; if only one does, you have built the wrong pair. And when $k = \pm 1$ the two constructions land on the same point, so report that angle once.

Step 2 is where marks are most often lost, and step 5 is where solutions go missing.

EXAMPLE 8.10

Solve $\cos x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

Take the reference angle from the size of k alone:

$$\hat{\theta} = \arccos \left| -\frac{1}{2} \right| = \arccos \frac{1}{2} = \frac{\pi}{3}.$$

Cosine is negative in the second and third quadrants, so build one angle in each:

$$\pi - \frac{\pi}{3} = \frac{2\pi}{3}, \quad \pi + \frac{\pi}{3} = \frac{4\pi}{3}.$$

Both lie in the interval, so $x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.

Notice that the calculator alone would have given only $\frac{2\pi}{3}$. The reference angle gives both, and the sign of k says where they live.

EXAMPLE 8.11

Solve $\sin x = -0.8$ for $0 \leq x \leq 2\pi$, giving the answers to three significant figures.

The constant is not one of the exact values, but the method does not change. The reference angle comes from the size of k : $\hat{\theta} = \arcsin 0.8 = 0.9273$. Sine is negative in the third and fourth quadrants, giving $\pi + 0.9273 = 4.0689$ and $2\pi - 0.9273 = 5.3559$. Both lie in the interval, so

$$x = 4.07 \quad \text{or} \quad x = 5.36 \quad (3 \text{ s.f.}).$$

Always ask which *other* quadrant shares the sign; that second angle is the one candidates most often miss.

TIP On a calculator paper, always check your solution count graphically: plot both sides and count the intersections in the interval. If your algebra found three solutions and the screen shows four, the screen is right.

When the argument is not plain x but $2x$ or $3(x-4)$, substitute for the whole argument and,

crucially, transform the interval along with it. A doubled argument sweeps the circle twice as fast, so it collects twice as many solutions.

EXAMPLE 8.12

Solve $\cos 2x = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$.

Let $u = 2x$. As x runs over $[0, 2\pi]$, u runs over $[0, 4\pi]$: two full turns.

The reference angle is $\hat{\theta} = \arccos \frac{\sqrt{3}}{2} = \frac{\pi}{6}$. Cosine is positive, so the first turn gives

$$u = \frac{\pi}{6} \quad \text{and} \quad u = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}.$$

The interval covers a second turn, so add 2π to each:

$$\frac{\pi}{6} + 2\pi = \frac{13\pi}{6}, \quad \frac{11\pi}{6} + 2\pi = \frac{23\pi}{6}.$$

A third turn would pass 4π , so the list is complete:

$$u = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6} \Rightarrow x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}.$$

Four solutions, not two. Note where the doubling acts: the step in u is still 2π , because $\cos u$ has period 2π whatever u came from. What doubling changes is the *interval*, and that is why two extra solutions appear. In x the period has become π .

8.6.1 When the interval is not 0 to 2π

The interval is part of the question, and it is not always $0 \leq x \leq 2\pi$. The commonest alternative is $-\pi \leq x \leq \pi$. That is still one whole turn, but measured differently: half a turn clockwise from the positive x -axis, and half a turn anticlockwise.

The method does not change. Build the two angles from the reference angle exactly as before, then shift either one by a whole turn if it falls outside the interval: subtract 2π from an angle above π , add 2π to an angle below $-\pi$. A whole turn returns the point to exactly where it was, so the value of the function is untouched.

EXAMPLE 8.13

Solve $\cos x = -\frac{1}{2}$ for $-\pi \leq x \leq \pi$.

The reference angle is $\hat{\theta} = \arccos \frac{1}{2} = \frac{\pi}{3}$. Cosine is negative, so the two quadrants are the second and the third:

$$\pi - \frac{\pi}{3} = \frac{2\pi}{3}, \quad \pi + \frac{\pi}{3} = \frac{4\pi}{3}.$$

The first already lies in $[-\pi, \pi]$. The second does not, because $\frac{4\pi}{3} > \pi$, so subtract one turn:

$$\frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}.$$

So $x = \frac{2\pi}{3}$ or $x = -\frac{2\pi}{3}$.

The two answers form a $\pm$ pair, and that is not a coincidence. Cosine is even, so on any interval symmetric about 0 its answers are $\pm\alpha$ for a single angle α . Sine is odd, and its pair on the same interval is

$$\hat{\theta} \text{ and } \pi - \hat{\theta} \quad \text{when } k > 0, \quad -\hat{\theta} \text{ and } -\pi + \hat{\theta} \quad \text{when } k < 0.$$

8.6.2 Using identities

An equation mixing $\sin 2x$ with $\sin x$, or $\cos 2x$ with $\cos x$, cannot be solved as it stands. An identity rewrites everything in terms of a single function, turning the equation into an algebraic one, usually a quadratic.

EXAMPLE 8.14

Solve $\cos 2x + 3 \cos x + 2 = 0$ for $0 \leq x \leq 2\pi$.

Choose the form of $\cos 2x$ that gives only cosines, $\cos 2x = 2 \cos^2 x - 1$:

$$2 \cos^2 x - 1 + 3 \cos x + 2 = 0 \implies 2 \cos^2 x + 3 \cos x + 1 = 0.$$

This factors as $(2 \cos x + 1)(\cos x + 1) = 0$, so $\cos x = -\frac{1}{2}$ or $\cos x = -1$. From the circle,

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{or} \quad x = \pi.$$

Picking the form of the double-angle identity that matches the other terms is the key step in these problems.

YOUR TURN 8.6 Solving Trigonometric Equations

21. Solve for $0 \leq x \leq 2\pi$.

(a) $\sin x = \frac{1}{2}$

(b) $\cos x = 0$

(c) $\tan x = 1$

(d) $\cos x = 1$

22. Solve for $0 \leq x \leq 2\pi$.

(a) $\sin x = -\frac{1}{2}$

(b) $2 \cos x = \sqrt{3}$

(c) $\sqrt{2} \sin x = -1$

(d) $\tan x = -\sqrt{3}$

23. Solve for $0 \leq x \leq 2\pi$. Transform the interval before solving.

(a) $\sin 2x = 0$

(b) $\cos 2x = \frac{1}{2}$

(c) $\tan 2x = 1$

24. Solve for $0 \leq x \leq 2\pi$, using an identity first.

(a) $2 \sin^2 x - \sin x = 0$

(b) $2 \cos^2 x + \cos x - 1 = 0$

(c) $\cos 2x + \sin x = 0$

(d) $2 \cos^2 x - 3 \cos x + 1 = 0$

(e) $\sin 2x = \cos x$

(f) $\tan^2 x - \tan x = 0$

25. Solve for $-\pi \leq x \leq \pi$. Give every answer inside that interval.

(a) $\cos x = \frac{\sqrt{3}}{2}$

(b) $\sin x = -\frac{1}{2}$

(c) $\tan x = -1$

(d) $2 \sin x = \sqrt{2}$

Chapter 8 · Trigonometry

Reflections

TOK A sine wave is periodic: it repeats forever, identical in every cycle. Yet it is built from the endlessly irrational π and takes irrational values at almost every point. How can something so regular be made of something so unruly? The physicist Joseph Fourier then proved that *any* repeating signal, however jagged, is a sum of pure sine waves. So the smooth circle is hidden inside every square wave and every spoken word. Is the sinusoid a human tool, or the alphabet in which nature writes all its rhythms?

IM Trigonometry began as astronomy. Babylonian and Greek observers needed to predict the wandering of the planets, and the sine was born as a tool for the geometry of the celestial sphere. Its greatest early development came in India and in the Islamic world. Astronomers there tabulated sines to many decimal places, to fix the times of prayer and the direction of Mecca. For two thousand years, the functions of this chapter were the mathematics of the heavens.

RW Because a sine wave is what steady rotation looks like from the side, it models every oscillation in science and engineering. Alternating current rises and falls as a sine while the generator spins. Sound is pressure oscillating in the same way, and its pitch is the frequency of the wave. Light, radio, and every signal your phone sends and receives are sinusoids too. Fourier analysis, decomposing a signal into its sine components, is how music is compressed, images are stored, and earthquakes and heartbeats are read.

EXT In radians, $\sin x$ and x almost agree for small x : $\sin 0.1 = 0.09983\dots$ On the unit circle x is the arc and $\sin x$ is the height, and over a tiny arc there is almost no difference between them. Physicists use this as the **small-angle approximation**: for $|x|$ small and in radians,

$$\sin x \approx x, \quad \tan x \approx x, \quad \cos x \approx 1 - \frac{1}{2}x^2.$$

At $x = 0.1$ these give 0.1, 0.1003 and 0.995, against true values 0.09983, 0.10033 and 0.99500. The pendulum equation is only solvable because $\sin \theta$ is replaced by θ , which is why a pendulum keeps time for small swings and drifts for large ones. All three fail in degrees, and that is what makes radians the natural unit for calculus. Chapter 13 turns the first into $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, the limit that gives $\frac{d}{dx} \sin x = \cos x$.

End-of-Chapter Problems — Chapter 8: Trigonometry

1. Use the unit circle to write down the exact value of each.

- | | | | |
|----------------------------|-----|---------------------------|-----|
| (a) $\sin \frac{2\pi}{3}$ | [1] | (b) $\cos \frac{2\pi}{3}$ | [1] |
| (c) $\tan \frac{2\pi}{3}$ | [2] | (d) $\cos \frac{5\pi}{4}$ | [2] |
| (e) $\sin \frac{11\pi}{6}$ | [2] | (f) $\tan \frac{3\pi}{4}$ | [2] |

2. The point P lies on the unit circle at angle θ .

- (a) State the coordinates of P when $\theta = \frac{5\pi}{6}$. [2]
- (b) P has coordinates $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$. Find θ for $0 \leq \theta \leq 2\pi$. [2]
- (c) P lies in the third quadrant. State the sign of $\sin \theta$, of $\cos \theta$ and of $\tan \theta$. [3]
- (d) Given $\sin \theta = \frac{\sqrt{3}}{2}$, write down the two possible values of θ in $0 \leq \theta \leq 2\pi$, and explain why there are exactly two. [3]

3. Prove each identity. They are arranged in increasing order of difficulty.

- (a) $\cos^2 \theta - \sin^2 \theta = 1 - 2\sin^2 \theta$ [2]
- (b) $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$ [3]
- (c) $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$ [3]
- (d) $\cos^4 \theta - \sin^4 \theta = \cos 2\theta$ [3]
- (e) $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$ [4]
- (f) $\frac{1}{\sec \theta - \tan \theta} = \sec \theta + \tan \theta$ [4]
- (g) $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$ [5]
- (h) $\frac{1 + \sin 2\theta}{\cos 2\theta} = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$ [5]

4. Given $\cos \theta = -\frac{3}{5}$ with θ in the second quadrant, find the exact value of each.

- | | | | |
|--------------------|-----|--------------------|-----|
| (a) $\sin \theta$ | [2] | (b) $\tan \theta$ | [1] |
| (c) $\sin 2\theta$ | [2] | (d) $\cos 2\theta$ | [2] |

5. Given $\tan \theta = 2$ with θ acute, find the exact value of each.

- (a) $\sin \theta$ and $\cos \theta$ [2] (b) $\sin 2\theta$ [2]
 (c) $\cos 2\theta$ [2]

6. Use a compound-angle formula to find each exact value.

- (a) $\tan 75^\circ$ [4] (b) $\cos 15^\circ$ [3]
 (c) $\sin 105^\circ$ [3] (d) $\tan 105^\circ$ [4]

7. Prove each result.

- (a) $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ [4]
 (b) $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ [5]
 (c) $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ [5]

8. Solve for $0 \leq x \leq 2\pi$.

- (a) $\sin x = \frac{1}{2}$ [2] (b) $\sqrt{3} \tan x = 1$ [3]
 (c) $2 \cos^2 x = 1$ [4] (d) $\tan^2 x = 3$ [4]

9. Solve for $0 \leq x \leq 2\pi$. Each becomes a quadratic.

- (a) $2 \sin^2 x + \sin x - 1 = 0$ [5] (b) $4 \sin^2 x - 3 = 0$ [4]
 (c) $\cos 2x = \cos x$ [5] (d) $\sin 2x = \sin x$ [5]

10. Solve for $0 \leq x \leq 2\pi$.

- (a) $2 \sin x \cos x = \frac{1}{2}$ [4]
 (b) $\cos 2x + 3 \sin x = 2$ [6]
 (c) $\sin x + \cos x = 1$ [6]

11. Solve each equation for $-\pi \leq x \leq \pi$. Build every answer from the reference angle, then shift by a whole turn where the interval requires it.

- (a) $\sin x = -\frac{1}{2}$ [3] (b) $\cos x = \frac{\sqrt{3}}{2}$ [3]
 (c) $\tan x = -1$ [3] (d) $2 \cos^2 x - 1 = 0$ [4]
 (e) $\sin 2x = \frac{\sqrt{3}}{2}$ [5] (f) $\tan\left(x + \frac{\pi}{4}\right) = \sqrt{3}$ [5]

12. Given the identity $1 + \tan^2 \theta = \sec^2 \theta$, solve $\sec^2 x = 4$ for $0 \leq x \leq \frac{\pi}{2}$. [4]

13. Evaluate, giving exact values.

- (a) $\arctan 1$ [1] (b) $\arcsin\left(-\frac{1}{2}\right)$ [1]

(c) $\arccos(-1)$ [1] (d) $\arcsin\left(\sin \frac{5\pi}{6}\right)$ [3]

14. Consider $f(x) = 3\sin(2x) + 1$.

- (a) State the amplitude and the period. [2]
(b) State the maximum and minimum values. [2]

15. The graph of $y = a\sin(bx) + d$ has maximum 7, minimum 1, and period π . Find a , b and d . [4]

16. Sketch $y = \cos x - 1$ for $0 \leq x \leq 2\pi$, stating the maximum, the minimum, and the intercepts. [4]

17. The depth of water in a harbour is $d(t) = 5 + 3\sin\left(\frac{\pi t}{6}\right)$ metres, t hours after midnight.

- (a) State the maximum and minimum depths. [2]
(b) Find the period, and say what it represents. [2]

18. A seat on a Ferris wheel has height $h(t) = 10 - 8\cos\left(\frac{\pi t}{15}\right)$ metres, t seconds after the start.

- (a) State the minimum and maximum heights. [2]
(b) Find the time for one full revolution. [2]

19. GDC Express $3\sin x + 4\cos x$ in the form $R\sin(x + \alpha)$ with $R > 0$.

- (a) Find R and α . [3]
(b) Hence state the maximum value. [1]

20. GDC Find the maximum value M of $5\sin x + 12\cos x$, and the value of x in $[0, 2\pi]$ at which it occurs. [5]

21. Use the symmetry of the unit circle to answer each part.

- (a) Given $\sin 0.6 = 0.565$, write down $\sin(\pi - 0.6)$. [2]
(b) Given $\cos 1.2 = 0.362$, write down $\cos(-1.2)$ and $\cos(2\pi - 1.2)$. [3]
(c) Explain why $\sin(\theta + 2\pi) = \sin \theta$ for every θ . [2]
(d) Explain why $\tan(\theta + \pi) = \tan \theta$, but $\sin(\theta + \pi) \neq \sin \theta$. [4]

22. Solve each equation on the interval given. Take care: the same equation has different answers on different intervals.

- (a) $\cos x = -\frac{\sqrt{2}}{2}$ for $0 \leq x \leq 2\pi$, and then for $-\pi \leq x \leq \pi$. [4]
- (b) $\sin x = -\frac{\sqrt{3}}{2}$ for $-\pi \leq x \leq \pi$. [3]
- (c) $\tan 2x = \sqrt{3}$ for $-\pi \leq x \leq \pi$. [5]
- (d) Explain why an equation of the form $\cos x = k$, with $|k| < 1$, always has two solutions on $-\pi \leq x \leq \pi$, and why they are negatives of each other. [3]

23. GDC Daylight in Sapporo. The number of hours of daylight is modelled by

$$L(t) = 12.2 + 3.4 \sin\left(\frac{2\pi(t-81)}{365}\right),$$

where t is the day of the year, with $t = 1$ on 1 January.

- (a) State the maximum and minimum daylight, and find the value of t at which each occurs. [4]
- (b) State the period of L , and say what it represents. [2]
- (c) Find the daylight on 1 January, to three significant figures. [2]
- (d) Find the two days of the year on which the daylight is exactly 12.2 hours, and explain what is special about them. [4]
- (e) Find the values of t for which $L(t) \geq 15$, and state how many days of the year have at least 15 hours of daylight. [6]

24. One function, two forms. Let $f(x) = 2 \sin x + \cos 2x$ for $0 \leq x \leq 2\pi$.

- (a) Show that $f(x) = -2 \sin^2 x + 2 \sin x + 1$. [3]
- (b) Let $u = \sin x$. Write f as a quadratic in u , and state the interval of values u can take. [3]
- (c) Hence show that the maximum value of f is $\frac{3}{2}$. [4]
- (d) Find the values of x in $[0, 2\pi]$ at which that maximum occurs. [3]
- (e) Solve $f(x) = 1$ for $0 \leq x \leq 2\pi$. [5]
- (f) State the range of f . [3]

25. Find the exact value where there is one, and write **undefined** where there is not.

- | | | | |
|--|-----|---|-----|
| (a) $\arcsin\left(\sin \frac{7\pi}{6}\right)$ | [2] | (b) $\arccos\left(\cos \frac{7\pi}{4}\right)$ | [2] |
| (c) $\arctan\left(\tan \frac{2\pi}{3}\right)$ | [2] | (d) $\sin\left(\arccos \frac{5}{13}\right)$ | [2] |
| (e) $\tan\left(\arcsin \frac{8}{17}\right)$ | [3] | (f) $\cos(\arctan 2)$ | [2] |
| (g) $\arccos(\sin \pi)$ | [2] | (h) $\sin(\arcsin 2)$ | [2] |
| (i) $\cos\left(\arcsin \frac{3}{5} + \arctan \frac{4}{3}\right)$ | [4] | (j) $\tan\left(2 \arctan \frac{1}{3}\right)$ | [3] |

26. Ten identities. Prove each one. They draw on the double-angle, compound-angle and reciprocal results of this chapter, and they are arranged in increasing order of difficulty.

- (a) $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$ [3]
- (b) $\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$ [3]
- (c) $\operatorname{cosec} 2\theta = \frac{1}{2} \sec \theta \operatorname{cosec} \theta$ [3]
- (d) $\frac{\sec \theta + 1}{\tan \theta} = \frac{\tan \theta}{\sec \theta - 1}$ [4]
- (e) $\tan\left(\theta + \frac{\pi}{4}\right) = \frac{1 + \tan \theta}{1 - \tan \theta}$ [4]
- (f) $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B$ [5]
- (g) $\frac{1 + \tan^2 \theta}{1 - \tan^2 \theta} = \sec 2\theta$ [5]
- (h) $\cot \theta - \tan \theta = 2 \cot 2\theta$ [5]
- (i) $\frac{\cos 2\theta}{1 + \sin 2\theta} = \frac{1 - \tan \theta}{1 + \tan \theta}$ [6]
- (j) $\operatorname{cosec} \theta - \cot \theta = \tan \frac{\theta}{2}$ [6]

Challenge Questions

27. GDC **The wave form, and when an equation can be solved.** A sum of a sine and a cosine of the same angle is itself a single shifted sine wave: $a \sin \theta + b \cos \theta = R \sin(\theta + \alpha)$. This investigation proves it and then uses it to decide, without solving, whether an equation has any solutions at all.

Throughout take $a > 0$ and $b > 0$.

- (a) Expand $R \sin(\theta + \alpha)$ and compare it with $a \sin \theta + b \cos \theta$ to show that $R \cos \alpha = a$ and $R \sin \alpha = b$. [3]
- (b) Hence show that $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$. (Hint: square and add) [3]
- (c) Write $7 \sin \theta + 24 \cos \theta$ in the form $R \sin(\theta + \alpha)$, and state its maximum and minimum values. [3]
- (d) Explain why the equation $a \sin \theta + b \cos \theta = k$ has a solution if and only if $|k| \leq R$. [4]
- (e) Solve $3 \sin x + 4 \cos x = 5$ for $0 \leq x \leq 2\pi$, giving your answer to three significant figures. Explain why this equation has exactly one solution in the interval. [5]

28. **$\cos n\theta$ as a polynomial in $\cos \theta$.** You have already met $\cos 2\theta = 2 \cos^2 \theta - 1$ and, in the end-of-chapter problems, $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. Each is a polynomial in $\cos \theta$ alone. This investigation shows that this always happens, and finds the pattern.

Write $c = \cos \theta$ throughout.

- (a) Use the compound-angle formulas to show that

$$\cos((n+1)\theta) = 2 \cos \theta \cos(n\theta) - \cos((n-1)\theta).$$

(Hint: expand $\cos(n\theta + \theta)$ and $\cos(n\theta - \theta)$, then add) [5]

- (b) Starting from $\cos 0 = 1$ and $\cos \theta = c$, use the recurrence to obtain $\cos 2\theta$ and $\cos 3\theta$ in terms of c . [3]
- (c) Use it again to find $\cos 4\theta$ and $\cos 5\theta$ in terms of c . [4]
- (d) State the degree of the polynomial for $\cos n\theta$, and the coefficient of its highest power. Justify both from the recurrence. [4]
- (e) Explain why no such polynomial in $\cos \theta$ alone exists for $\sin n\theta$ when n is even. (Hint: compare $\sin(-\theta)$ with $\sin \theta$) [3]

29. GDC **The window of safe passage.** At a harbour mouth the depth of water is modelled by

$$d(t) = 6 + 2 \sin\left(\frac{\pi t}{6}\right) \text{ metres,}$$

where t is the number of hours after midnight. A ferry can enter only while the depth is at least the figure its hull requires. This investigation finds the window of time available, and then finds it in general.

- (a) State the greatest and least depths, the period of the tide, and the time of the first high tide. [4]
- (b) The ferry needs at least 7 m of water. Solve $d(t) \geq 7$ on $0 \leq t \leq 12$, and state the window as a pair of clock times. [5]
- (c) Show that the window is exactly 4 hours long, and explain why the same window opens once in every tidal cycle. [3]
- (d) Now take a general tide $d(t) = m + a \sin\left(\frac{2\pi t}{T}\right)$ and a required depth k with $m < k < m + a$. Show that the time available in each cycle is

$$\frac{T}{\pi} \arccos\left(\frac{k - m}{a}\right).$$

(Hint: solve for the two ends in terms of $\arcsin$, subtract, then use $\frac{\pi}{2} - \arcsin c = \arccos c$) [6]

- (e) A larger ferry needs 7.5 m. Use the formula to find its window, to the nearest minute. Then explain, from the shape of the arccosine graph, why the last half-metre of clearance costs far more time than the first. [5]

